

VR-Forms — v1.0.3 (revision note)

Two editorial corrections: synchronisation with VR v1.1.1, and scoping of the “no ontology” language

Vitaly Reznik

2026 — Version 1.0.3 (revision note)

Scope. This is the revision delta for VR-Forms v1.0.3. *No construction, definition, theorem, register, or axiom-profile of VR-Forms is altered.* Two editorial changes: (1) the recall of the parent work VR is brought up to date with VR v1.1.1; (2) the “no ontology” language is scoped to “no ontology of substance”.

(1) Correction: the parent-VR recall

The introduction described VR as

“a minimal axiomatisation of arithmetic on **three primitives** $\{\emptyset, \rightarrow, t\}$ and **four axioms**; equivalent to Peano arithmetic.”

This is superseded. In VR v1.1.1 the propositional implication $\rightarrow$ (and the principles A1, A2 and the two-valued logical register) are **removed**: $\rightarrow$ generated only a finite truth-function algebra used by no VR theorem — arithmetic, not logic — and VR generates no logic of its own (the logic it reasons with is metatheoretic, borrowed). The corrected recall reads

“a minimal *axiom-free* reconstruction of arithmetic on **two primitives** $\{\emptyset, t\}$ — the nullary base $\emptyset$ and the unary succession t — with no axioms of its own (membership is defined; the succession principle A3 is a theorem; induction A4 is the recursor); equivalent to first-order Peano arithmetic (Zenodo DOI 10.5281/zenodo.20688164, v1.1.1; concept 10.5281/zenodo.20212091).”

(2) Correction: scoping the “no ontology” language

Several places in VR-Forms state, without qualification, that “the cycle ascribes no ontology” (and the “ontological register” label is dropped on that basis). Read at face value this is a *blanket* denial of ontology, and it collides with the global membership relation $\in$ that VR-Sets does carry. The intended and accurate claim is narrower: VR ascribes no ontology *of substance* — no object that *is* prior to operations; being is relocated into doing. It is not the claim that the cycle ascribes no ontology *at all*: the operational/set layer carries genuine relational and structural commitments (global $\in$, potentialist infinity, relational identity), which are *tracked* (the two registers and the axiom tiers), not denied. Accordingly every unqualified “ascribes no ontology” should read “ascribes no ontology *of substance*”; the disclaimer, where it belongs, localises to the *formal* register (“syntactic descriptions without ontological commitment”), not to the cycle as a whole.

Why VR-Forms is otherwise unaffected

VR-Forms (the operational two-register apparatus over VR-Sets, with the conservativity and transit results) does not depend on the removed layer: its content references VR only through the operational reading and the nullary base $\emptyset$, which is unchanged in v1.1.1. The Lean module (`VRCycle.Forms`) contains no reference to the removed objects (`VRBool`, `impl`, `A1`, `A2`), and the build is unaffected. The “propositional floor” of the conservativity theorem (`Forms/Conservativity.lean`) is an internal technical layer of Forms, unrelated to VR’s former $\rightarrow$ operator.

Prepared with the assistance of Claude (Anthropic, Opus 4.8) in the cycle’s Variant A architecture; all mathematical content and positions are due to the author, Vitaly Reznik.